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Correlation between PEAK relational training system and one-word picture vocabulary tests

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ABSTRACT

The past few decades of research in autism spectrum disorders have been successful in developing effective behavioral treatments; however, the psychometrics of these strategies has not been documented well in applied settings. The current experiment evaluated the relationship between established measures of language skills (receptive and expressive one-word picture vocabulary test; ROWPVT-4 and EOWPVT-4, respectively) and a recently released assessment and curriculum tool designed to teach instructional skills using a behavior analytic approach (promoting the emergence of advanced knowledge relational training system; PEAK). Each participant was administered three assessments: The PEAK direct training module assessment, the ROWPVT-4 assessment, and the EOWPVT-4 assessment. Scores from all three assessments were compared to assess the relationship between each assessment. The results indicated both a strong correlation between the PEAK direct training module and commonly used language assessments (ROWPVT-4 and EOWPVT-4), as well as strong reliability in the administration of the assessments.

1. Introduction

Developmental disabilities and autism spectrum disorder affect a large number of families in the world today. According to the Center for Disease Control's Autism and Developmental Disabilities Monitoring Network Surveillance Year 2008 Principal Investigators (2012), 1 in 68 children currently have a diagnosis of autism spectrum disorder. This study also showed that 1 in 6 children currently have a diagnosis of a developmental disability. These findings combined with previous years' findings indicate a trend of increasing diagnoses for all developmental disabilities. This increasing trend in diagnoses highlights the need for effective treatments to maximize these individuals' potential. As might be expected, the increasing number of diagnoses of individuals with developmental disabilities has led to an increasing number of individuals who require special support within the education system. The U.S. Department of Education, National Center for Education Statistics (2011) investigated the trends in the number of students with disabilities who qualify for specialized support in public schools. From the 1998–1999 school year to the 2008–2009 school year, the percentage of students enrolled in public schools in the United States with a diagnosis of autism rose from 0.1% to 0.7%. These students represent only a portion of the students who are included in the Individuals With Disabilities Education Act (IDEA) (2004) which governs the services of children with disabilities. The findings of the NCES (2011) indicate that as the percentage of all students with developmental

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disabilities increased from 8.3% to 13.2% in the same 10 year period. In response to the growing number of students with special needs in the education system, it is vital to continually evaluate and improve the treatments and services that are available to aid in special needs instruction.

Among the currently utilized treatments for individuals with developmental disabilities, those based on the principles of applied behavior analysis (ABA) are the most widely used (Hess, Morrier, Heflin, & Ivey, 2008; National Autism Center, 2009). Treatments based in ABA such as early intensive behavior intervention (EIBI) and discrete trial training (DTT) are among the most frequently used and have been demonstrated to result in increases in academic achievement, adaptive behavior skills, and social skills (Myers & Johnson, 2007) as well as specific verbal and language skills (e.g., Greer, Yaun, & Gautreaux, 2005; Yamamoto & Mochizuki, 1988). Furthermore, the implementation of these treatments is not restricted to trained behavior analysts. Previous research has demonstrated successful training of DTT with both staff and teachers as well. In one study incorporating behavioral skills training, three teachers were effectively taught to conduct DTT by giving instructions, feedback, rehearsing, and modeling the procedure (Sarakoff & Sturmey, 2004). A second study by Thomson et al. (2012) trained tutors to conduct DTT with children with autism. Tutors were initially taught via a written manual. If they failed to reach the 80% mastery criterion, they were instructed to watch a video that modeled DTT. Each tutor was able to meet the criterion to teach DTT following these procedures. With the research supporting the utility of ABA in school and professional contexts, its incorporation into assessment and curriculum tools is warranted.

One such assessment and curriculum tool was recently released: The promoting the emergence of advanced knowledge relational training system (PEAK; Dixon, 2014a,b). The PEAK is comprised of four separate modules, each addressing 184 targeted skills of increasing difficulty: direct training, generalization, equivalence, and transformation of functions. All four modules incorporate ABA principles in training skills which are identified by assessments included with each module. Each module presents the instructor with the steps and modalities with which to teach each skill. Where the PEAK is the most comparable to currently available treatments or instructional protocols for individuals with developmental disabilities is in the direct training module (Dixon, 2014a). This module incorporates DTT instruction including principles of reinforcement and prompt and correction techniques for individual responses during instruction delivery. While the PEAK is similar to other currently available assessment and curriculum tools, it differs in a several meaningful ways. First, the PEAK not only provides an assessment tool and a curriculum tool, it also provides comprehensive instructions on the methods of teaching the skills, and a ready-made data collection system allowing for use by novice instructors. Second, the PEAK is the only currently available ABA-based curriculum and assessment tool with empirical demonstration of its reliability and validity. Previous research on the direct training module (Dixon, 2014a) has indicated a strong criterion validity with established language and academic assessments (Dixon et al., 2014a), and with established measures of intelligence quotients (Dixon, Whiting, Rowsey, & Belisle, 2014b). Furthermore, both previously published studies indicated high rates of inter-rater reliability on the PEAK direct training assessment tool. With the need for empirically validated treatments for individuals with developmental disabilities, it is important to continue to investigate the utility of our current tools to ensure the most effective treatment possible. Specifically, research should continue to look at specific areas of deficit in individuals with disabilities and how our currently available tools address or fail to address the needs.

One area in which individuals with disabilities frequently display deficits is in language skills. One of the most commonly used language tests is the one-word picture vocabulary test—fourth edition. This test consists of two parts including the receptive one-word picture vocabulary test (ROWPVT-4; Martin & Brownell, 2011b) and the expressive one-word picture vocabulary test (EOWPVT-4; Martin & Brownell, 2011a) used together to assess language skills. The test was developed as a standardized test to assess an individual's language skill repertoire, as well as evaluate the individual's language skills compared to the language skills of typically developing peers of the same age. The assessment can be efficiently administered in about 15–20 min per child and covers an age range from 2 to 95 years old. The current study sought to replicate and extend previous research on the PEAK by investigating its relationship to the ROWPVT-4 and the EOWPVT-4 as well as further investigating the reliability of the PEAK assessment as used to evaluate individuals with autism and other developmental disabilities.

2. Materials and methods

2.1. Participants and setting

Forty participants were recruited from a school serving individuals with autism and other related disabilities in the Midwest. All participants had been receiving several forms of treatment services for a minimum of three months. These therapies included speech and language, art, music, and behavior therapy, etc. 13 participants were removed from the analysis either due to severe problem behavior which precluded an accurate assessment of their skills, or due to a perfect score on an assessment which may limit the accuracy of the comparisons between scores. The remaining participants included 27 students between the ages of 5 and 22 ($M = 13.33$, $SD = 4.81$) years who carried a diagnosis of some form of developmental disability (see Table 1).

All evaluations were conducted either in the participant's classroom or in the library at the participant's school. During sessions conducted in the school library, the student and one or two assessors were the only individuals present. For sessions conducted in the student's classroom, a teacher, one or two aides, and three to six other students were present in the classroom. To minimize disturbances of either the classroom activities or the assessment itself, during the assessment the